Senior AI & Automation Engineer – n8n-EKS- UpGuard – 19/8/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI & Automation Engineer/GenAI /Integration Solution Architect/Full Stack Developer/Senior DevOps /Data Engineer/ Software Engineer/Mulesoft /AI/ML/DL/BI/SharePoint/Dynamics 365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior AI & Automation Engineer/Python/AI/ML/DL/SP/D365 Developer/Snr Senior DevOps/Data Engineer
Responsibilities:

- **Automation and Workflows**
- Developed and deployed critical n8n workflows in a hybrid cloud environment, seamlessly integrating services across AWS and GCP. Orchestrated complex processes that connect platforms like Gong (customer call analysis), Notion (knowledge management), and BigQuery (data warehousing) to democratize data and insights, ensuring enterprise-grade reliability and compliance. aws eks --region ap-southeast-2 update-kubeconfig --name n8n-cluster
- Designed scalable infrastructure, secure credential management, and production-ready automation workflows that drive operational efficiency and informed decision-making.
- Designed and deployed scalable n8n workflows on a hybrid cloud infrastructure (AWS & GCP) to automate and optimize processes for hundreds of UpGuard team members. kubectl scale deployment/n8n-worker --replicas=5
- Orchestrated critical automations for data ingestion, reporting, and internal operations, effectively democratizing data access and improving team productivity. n8n.run_workflow('Gong -> AI-Summary -> Notion')
- Implemented robust governance and security controls within the n8n platform to ensure enterprise-grade reliability and compliance as the organization scaled.
  export N8N_CREDENTIALS_ENCRYPTION_KEY=$(aws secretsmanager get-secret-value --secret-id n8n_prod_key)
- Spearheaded the democratization of n8n usage by creating comprehensive templates, documentation, and training materials, enabling non-technical teams to autonomously build and deploy automation workflows.
- Architected and implemented n8n workflows to integrate with the UpGuard platform, creating reusable templates for critical functions such as automated vendor onboarding, attack surface issue remediation, and data leak alerts.
- Architected and implemented a code-based governance layer within n8n workflows to enforce privacy and compliance, including logic (if ($json.containsPII)) to automatically stop processing and flag workflows that handle Personally Identifiable Information (PII) for review, proactively preventing data breaches and non-compliant activity.
- Successfully implemented and managed scalable MLOps and LLMOps solutions on AWS, driving efficiency by using SageMaker for streamlined model operationalization and Amazon Bedrock to build and scale generative AI applications.
- **AI and Machine Learning**
- Develop and deploy domain-specific LLMs to enhance UpGuarder productivity, including a Vendor Risk Assessment Co-pilot for automated questionnaire analysis, a Threat & Vulnerability Explainer for simplified risk communication, and a Compliance & Policy Drafter for accelerated policy creation.
- Fine-tuned lightweight models on internal data from sources like Gong (customer call analysis), Notion (knowledge management), and BigQuery (data warehousing), and deployed them via intuitive interfaces—such as internal dashboards, Slack bots, and custom APIs—to empower teams across the organization with data-driven insights.
- Architected and implemented a code-based governance layer within n8n workflows to enforce privacy, compliance, and cost controls, including logic (if ($json.llmCost > 500.00)) to automatically stop processing, flag workflows that exceed a daily budget threshold, or handle Personally Identifiable Information (PII) for review.
- Leveraged PyTorch, TensorFlow, and Hugging Face Transformers to develop and deploy LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and deep learning models, including fine-tuning pre-trained models for NLP and analytics.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs).
- Integrated MongoDB Atlas as a vector database for RAG, enabling real-time and context-aware enterprise insights while implementing security measures to prevent data loss from traditional JavaScript and new AI prompt injection attacks.
- Implemented cloud-native architectures with ECS/EKS for containerized microservices & REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within an Amazon Redshift-powered lakehouse.
- Democratised n8n usage by designing reusable templates, documentation, & training resources, empowering non-technical teams to independently build & deploy automation workflows with confidence — including safeguards such as automatically flagging workflows that exceed daily budget thresholds or routing Personally Identifiable Information (PII) for review
- **Embed security and governance**

- Bake privacy, compliance, and cost controls into every workflow and model deployment, ensuring enterprise-grade reliability and data protection. if ($json.containsPII) { throw new Error('PII data detected. Workflow stopped for manual review.');…
- Enabled widespread adoption of AI and automation by conducting workshops, providing support, and mentoring team members on leveraging these technologies effectively.
- Integrated with Power Platform (Power Automate, Power Apps, Power BI) by designing & developing Custom Connectors for secure API interaction (via Azure API Management, Azure API for FHIR, Azure Container Apps, Azure Spring Apps), and enhanced automation with AI Builder. This included developing Power Apps Component Framework (PCF) controls for specific and enhanced visualization of fetched external CRM data within the model-driven application interface.
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Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Snr AI & Automation Engineer / Python/AI/ML/DL/SP/D365 Developer/Snr Senior DevOps/Data Engineer
Responsibilities:

- **Automation and Machine Learning:** Orchestrated ML/DL pipelines in a hybrid cloud environment (AWS & GCP) to automate and optimize high-frequency trading and risk management processes.
- Engineered and deployed scalable deep learning models for predictive analytics and algorithmic trading, integrating key data sources to generate real-time market insights. The models leveraged a distributed architecture to handle terabytes of historical market data.
- Designed and implemented a data ingestion and processing framework using custom machine learning pipelines, replacing manual workflows to democratize access to critical trading data. This framework allowed for real-time analysis and informed decision-making across the firm.
- Pioneered a code-based governance and security layer within the ML/DL workflows. The system included proactive PII detection and automated flagging to prevent data privacy breaches, ensuring compliance with strict financial regulations.
- Spearheaded the integration of machine learning models into a real-time trading environment, creating reusable components for critical functions such as automated sentiment analysis, anomaly detection for fraud prevention, and real-time market signal processing.
- Developed and delivered comprehensive training programs and documentation to upskill trading and operations teams on the use of new ML and DL tools, fostering a culture of data-driven decision-making and self-service analytics.
- Implemented robust security and credential management systems to ensure enterprise-grade reliability and compliance, protecting sensitive financial data and intellectual property within a hybrid cloud infrastructure.
- Built robust predictive models and secure machine learning APIs using diverse algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests, and SVM) and machine learning paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and TensorFlow to drive propensity modeling and pattern discovery.
- Built and exposed robust RESTful APIs for real-time data exchange across core banking systems and regulatory gateways.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.
    1. API Development: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
    2. System Integration: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.
- Engineered scalable, mission-critical backend services and production applications in Python, driving robust system integrations and automated deployments with code such as requests.post(f"{api_url}/v2/deployments", json={'app':'service', 'version':'v1.0'}).
- **AI & Machine Learning:** Developed and deployed domain-specific machine learning models to enhance trading and risk management, including models for automated sentiment analysis, predictive market movement, & real-time fraud detection.
- Fine-tuned and deployed lightweight models on internal data from sources like historical trade logs, market feeds, and internal knowledge bases. Models were made accessible via internal dashboards and APIs to provide real-time, data-driven insights to trading teams.
- Leveraged PyTorch and TensorFlow to develop and deploy deep learning models, including fine-tuning pre-trained neural networks for financial time-series analysis and natural language processing. Built robust predictive models and secure APIs using diverse machine learning algorithms (e.g., Random Forest, Gradient Boosting, SVM) and paradigms (supervised, unsupervised), applying Scikit-learn to drive next-best-action recommendations and detect complex trading patterns.
- Designed and maintained scalable AWS infrastructure (EC2, S3, RDS, Lambda) for hosting advanced analytics pipelines and machine learning models, ensuring a reliable and high-performance environment for financial operations.
- **Data Governance and Democratization:**
- Developed a Python-based data analysis framework leveraging Pandas and Jupyter Notebooks, creating reusable templates and documentation to empower non-technical teams with self-service analytics and reporting.
- Embedded security and governance into every data pipeline and model deployment, ensuring enterprise-grade reliability and strict data protection. We implemented automated logic to flag and stop any process handling sensitive financial data, preventing potential breaches and ensuring compliance with regulatory standards.
- Enabled widespread adoption of new machine learning tools by conducting workshops, providing ongoing support, and mentoring team members on leveraging these technologies effectively for market analysis and risk management.

- **Python Applications**: Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for automated data indexing and natural language processing. Integrated custom speech-to-text APIs for the transcription and sentiment analysis of internal communications, directly logging key insights into our CRM and trade logs.
- **Deep Learning Models**: Leveraged Theano and TensorFlow to develop and deploy deep learning models, including fine-tuning pre-trained models for financial time-series analysis and market sentiment.
- Integrated MongoDB Atlas as a vector database for a semantic search engine to enable real-time and context-aware enterprise insights, successfully preventing data loss from traditional JavaScript and AI prompt injection attacks.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr ML & Automation Engineer/Integration Solution Architect/Py/ML/Mulesoft Developer/DevOps Engineer
Responsibilities:

- **Process Automation and Predictive Analytics**
- Orchestrated automated data pipelines using on-premise servers and early cloud services to optimize international student enrolment and admission processes.
- Engineered and deployed predictive models for analytics and student profiling, integrating key data sources to generate real-time insights on applicant eligibility and success rates. Models leveraged a distributed architecture to handle large volumes of application data.
- Designed and implemented a data ingestion and processing framework using custom algorithms and scripts to replace manual workflows. This democratized access to critical student data, enabling real-time analysis and informed decision-making across the admissions department.
- Pioneered a code-based governance and security layer within data processing workflows. The system included proactive PII detection and automated flagging to prevent student data privacy breaches, ensuring compliance with strict educational regulations.
- Spearheaded the integration of predictive models into the student information system, creating reusable components for critical functions such as automated document verification, fraud detection, and real-time application status updates.
- Developed and delivered comprehensive training programs and documentation to upskill admissions and administrative teams on the use of new analytics tools, fostering a culture of data-driven decision-making and self-service reporting.
- Implemented robust security and credential management systems to ensure enterprise-grade reliability and compliance, protecting sensitive student data and intellectual property within the university's infrastructure.
- Built robust predictive models and secure APIs using diverse machine learning algorithms (SVM, KNN, ANN, regression, decision trees, random forests, and clustering). Applied frameworks like Scikit-learn to analyze application data and drive propensity modeling for student retention and course recommendation.
- AI & Predictive Analytics
- Developed and deployed domain-specific machine learning models to enhance student enrolment and administrative processes, including models for automated sentiment analysis of applications, predictive analytics for student retention, and real-time fraud detection. Fine-tuned and deployed lightweight models on internal data from sources like historical application forms and internal knowledge bases. Models were made accessible via internal dashboards and APIs to provide real-time, data-driven insights to admissions teams.
- Architected and implemented a code-based governance layer within data processing workflows to enforce privacy and compliance. Included logic to automatically flag or stop processing any data that exceeded predefined thresholds or contained sensitive student information, proactively preventing privacy breaches. Leveraged **Theano** and **TensorFlow** to develop and deploy deep learning models, including fine-tuning pre-trained neural networks for text analysis and student behavior patterns.
- Built robust predictive models and secure APIs using diverse machine learning algorithms (e.g., **Random Forest**, **Gradient Boosting**, **SVM**) and paradigms (**supervised**, **unsupervised**). Models applied **Scikit-learn** to analyze application data and detect complex patterns. Designed and maintained scalable **AWS infrastructure (EC2, S3, RDS)** for hosting advanced analytics pipelines and machine learning models, ensuring a reliable and high-performance environment for university operations. Implemented cloud-native architectures with **AWS EC2/ECS** for containerized microservices and **REST APIs**, ensuring efficient model integration with our student information system and streamlined data ingestion within an institutional data warehouse.
- **Process Automation and AI Democratization**
- Developed a Python-based data analysis framework leveraging Pandas and Jupyter Notebooks, creating reusable templates and documentation to empower non-technical teams with self-service analytics and reporting.
- Embedded security and governance into every data pipeline and model deployment, ensuring enterprise-grade reliability and strict data protection. Implemented automated logic to flag and stop any process handling sensitive data, preventing potential breaches and ensuring compliance with regulatory standards.
- Enabled widespread adoption of new machine learning tools by conducting workshops, providing ongoing support, and mentoring team members on leveraging these technologies effectively.
- **AI and Machine Learning**

- Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for automated data indexing and natural language processing.
- Leveraged Theano and TensorFlow to develop and deploy deep learning models, including fine-tuning pre-trained models for text analysis.
- Integrated MongoDB Atlas as a vector database for a knowledge graph to enable real-time and context-aware enterprise insights, successfully preventing data loss from traditional JavaScript and AI prompt injection attacks.
- Implemented cloud-native architectures with AWS EC2/ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion within an institutional data warehouse.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: ML & Automation Engineer / Software Test Engineer / DevOps / Data Engineer
Responsibilities:

- **Application Development & Machine Learning**
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Data Engineering & Architecture:
- Architected and implemented data solutions using Amazon Web Services (AWS), focusing on traditional data warehousing and ETL processes.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements, creating mapping documents that defined how source data fields corresponded to target data fields.
- Automation & DevOps:
- Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.
- Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- **Software Development and Deployment**
- Developed and deployed scalable web applications using Python and deployed them on traditional servers, a foundational skill for building production systems.
- Implemented APIs and integrated with diverse systems, ensuring seamless data exchange and system interoperability.
- Automation & Data Engineering:
- Managed server infrastructure using shell scripts and other automation tools, (e.g., Puppet, Chef).
- Integrated automation scripts into CI/CD pipelines (e.g., Jenkins) to ensure continuous and secure application delivery.
- Utilized relational databases for data storage and retrieval, demonstrating strong data management fundamentals essential for working with internal data sources.
- Security and Governance:
- Implemented security best practices, including regularly patching systems and applications, to ensure system integrity and compliance.
- Monitored server resources and implemented logging and security measures, a crucial skill for embedding security and governance into automation workflows.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- **Automation and CI/CD**:
- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Used scripting languages like Bash and configuration management tools to automate deployment processes and enhance scalability.

- **Infrastructure Management and Security**:
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Ensured robust security measures were in place to safeguard the infrastructure, demonstrating a foundational understanding of data protection and compliance.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- **System Modernization and Automation**:
- Served as Project Manager to replace the existing insurance web application with a new system, significantly reducing IT operational costs and improving system efficiency.
- Planned, implemented, and supported core IT systems including VMware ESX & vSphere 4.x, and applied security principles and software management best practices.
- **System Administration and Support**:
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support for system-related issues, demonstrating an ability to troubleshoot and resolve technical challenges.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), LM-Studio, Azure OpenAI ML Services, Copilot, Google GEMINI, MakerSuite

- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
  Workflow & Automation Platforms: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Senior AI Engineer – EKS- Reapit – 13/8/2025 – Failed on 20/8/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior AI Engineer /GenAI /Automation Engineer /Enterprise Innovation /Full Stack Developer/Snr DevOps /Data Engineer / Software Engineer/Mulesoft /AI/ML/DL/BI/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr AI Engineer/Automation Engineer /Python/AI/ML/DL/SP/D365 Developer/Snr Senior DevOps/Data Engineer
Responsibilities:

- **Agentic Platform Development**
- Architected and implemented AI agent architecture using LangChain/TensorFlow to build self-learning systems that adapt based on customer transaction patterns, significantly increasing automation and system performance.
- Developed advanced agentic applications leveraging cutting-edge agentic frameworks including LangGraph, AutoGen, CrewAI, and Semantic Kernel for sophisticated LLM orchestration and autonomous AI agent development.
- Utilized inter-agent communication protocols and seamless integration with external tools (e.g., web search APIs) within these agentic applications, further operationalizing through n8n for workflow automation to drive enhanced operational efficiency, automation, and intelligent decision-making.
- AI-Augmented RPA with Natural Language Processing (NLP): Engineered intelligent RPA workflows powered by NLP for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- AI-Augmented RPA with Computer Vision: Developed computer vision-based autonomous agents to integrate image recognition for the automated validation of patients' property documents and anomaly detection, streamlining regulatory workflows and enhancing decision-making accuracy.
- Developed computer vision-based autonomous agents integrating image recognition to automate property document validation and anomaly detection within agency workflows.
- Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for AI services with core code such as `requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'})`.
- **AI-Powered Automation & Insights**
- Led end-to-end GenAI solution development from secure audio ingestion, automated transcription, LLM orchestration, RAG pipeline integration, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- Developed scalable Python applications, orchestrating LangChain and LlamaIndex for LLM orchestration and data indexing. Integrated Amazon Transcribe Call Analytics for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Developed scalable Python applications, leveraging LangChain (for LLM orchestration) and LlamaIndex (for data indexing and retrieval), Whisper for accurate transcription of high-volume customer calls, Bedrock for accessing foundational models, and OpenAI frameworks, to integrate GenAI models with real-world data and CRM workflows, and deploying and optimising these systems on AWS while integrating with APIs, microservices, and data platforms.
- Developed scalable Python applications, leveraging Generative AI frameworks (LangChain for LLM orchestration, LlamaIndex for RAG) and speech-to-text models like Whisper to automate the transcription, summarization, and analysis of customer calls. Optimized deployment of these systems on AWS (using Bedrock and OpenAI frameworks) while integrating with APIs, microservices, and enterprise data platforms.
- Leveraged PyTorch, TensorFlow, and Hugging Face Transformers to develop and deploy LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and deep learning models, including fine-tuning pre-trained models for NLP and analytics.

- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs).
- Integrated MongoDB Atlas as a vector database for RAG (Retrieval-Augmented Generation), enabling real-time and context-aware enterprise insights.
- Implemented cloud-native architectures with AWS ECS/EKS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within an Amazon Redshift-powered lakehouse.
- Developed an intelligent AI agent using an Amazon Scraper (Python, requests, BeautifulSoup, pandas) with Streamlit UI, demonstrating robust data acquisition and constraints bypassing.
- **Technical Leadership & Innovation**
- Leveraged a self-recursive Artificial Superintelligence Architecture (ASI-ARCH) on the Harrison Platform (Nvidia DGX-1 systems) to autonomously innovate and optimize AI models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition.
- Utilized ASI-ARCH (Artificial Superintelligence Architecture) on Nvidia DGX-1 systems for autonomous innovation in AI models for financial management, enhancing algorithmic trading strategies and risk mitigation.
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimized data pipelines for AI model inference, enabling real-time predictions on large datasets.
- Automated the deployment of scalable GenAI applications and microservices across hybrid cloud environments, leveraging cloud-native AWS services like ECS/EKS, Lambda, & SageMaker to ensure high availability & streamlined CI/CD pipelines.
- Championed Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (GitHub Actions) for secure, continuous GenAI application delivery across complex hybrid cloud environments.
- Automated creation of secure, scalable IaC CI/CD pipelines and isolated GenAI platforms using LLMs (OLLLAMA4.0 on private clouds), MCP+RAG architectures, and agentic n8n workflows, enabling robust deployment of Python (Django, Flask) and Next.js server-side applications across AWS, Azure, and GCP.
- Integrated with Power Platform (Power Automate, Power Apps, Power BI) by designing & developing Custom Connectors for secure API interaction (via Azure API Management, Azure API for FHIR, Azure Container Apps, Azure Spring Apps), and enhanced automation with AI Builder. This included developing Power Apps Component Framework (PCF) controls for specific and enhanced visualization of fetched external CRM data within the model-driven application interface.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior AI/ML Engineer/Automation Engineer/Snr DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- **Agentic Platform Development**
- Architected and implemented multi-agent systems using Python to build self-learning systems that adapted based on customer transaction patterns, significantly increasing automation and system performance.
- Engineered intelligent rule-based RPA workflows powered by NLP libraries (e.g., NLTK, spaCy) for automated lead qualification and customer interaction analysis, improving data-driven decision-making.
- Developed computer vision models (e.g., using OpenCV and Scikit-image) to integrate image recognition for the automated validation of property documents and anomaly detection, streamlining regulatory workflows and enhancing decision-making accuracy.
- Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for machine learning services with core code such as requests.post(f"{api_url}/api/v1/deployments", json={'app':'ml-service', 'version':'v1.0.0'}).
- **ML/DL-Powered Automation & Insights**
- AI Solution Development: Led end-to-end machine learning solution development from secure audio ingestion, automated transcription, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- Python Applications: Developed scalable Python applications, leveraging frameworks like scikit-learn and NLTK for data indexing and natural language processing. Integrated speech-to-text APIs (like early AWS Transcribe, though this was limited at the time, or other third-party services) for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Deep Learning Models: Leveraged PyTorch and TensorFlow to develop and deploy deep learning models, including fine-tuning pre-trained models for NLP and analytics.
- Predictive Models: Built robust predictive models and secure machine learning APIs using diverse algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive next best action (NBA) recommendations and pattern discovery.
- AWS Infrastructure: Used AWS infrastructure (EC2, S3, RDS, Lambda, SageMaker) to design, build, and maintain scalable, secure systems for hosting advanced machine learning models.

- Data Integration: Integrated databases like PostgreSQL or MongoDB for real-time and context-aware enterprise insights.
- Cloud-Native Architectures: Implemented cloud-native architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within a data lakehouse.
- Web Scraping: Developed an intelligent agent using an Amazon Scraper (Python, requests, BeautifulSoup, pandas) to demonstrate robust data acquisition and constraints bypassing, with a simple UI built using frameworks like Django or Flask.
- **Technical Leadership & Innovation**
- Utilized Nvidia DGX-1 systems to develop and optimize machine learning models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering new market principles beyond human intuition.
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimizing data pipelines for machine learning model inference and enabling real-time predictions on large datasets.
- Automated the deployment of scalable machine learning applications and microservices across hybrid cloud environments, leveraging cloud-native AWS services like EC2, ECS, and Lambda to ensure high availability and streamlined CI/CD pipelines.
- Championed Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (Jenkins, Bamboo, or other build automation tools) for secure, continuous machine learning application delivery across complex hybrid cloud environments.
- Automated the creation of secure, scalable IaC CI/CD pipelines and isolated machine learning platforms, enabling robust deployment of Python (Django, Flask) and other server-side applications across AWS, Azure, and GCP.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior ML Engineer/Automation Engineer/Snr DevOps/Data Engineer /ML/DL/Mulesoft Developer
Responsibilities:

- **Automated Systems & Machine Learning**
- Architected and implemented multi-agent systems using Python to build self-learning systems for projects that adapted based on data patterns, significantly increasing automation and system performance.
- Engineered intelligent rule-based workflows powered by NLP libraries (e.g., NLTK) for automated data processing and interaction analysis, improving data-driven decision-making.
- Developed computer vision models (using OpenCV and Scikit-image) to integrate image recognition for the automated analysis of visual data and anomaly detection, streamlining data processing and enhancing decision-making accuracy.
- **Machine Learning & Data Insights**
- Led end-to-end machine learning project development, from data collection and automated transcription to statistical analysis and intelligent summarization, ensuring efficient data flow for research and academic projects.
- Developed scalable Python applications, leveraging core frameworks like scikit-learn and NLTK for data preprocessing, modeling, and natural language processing. Utilized third-party speech-to-text services for automatic transcription and sentiment analysis, integrating results into custom-built data analysis pipelines.
- Leveraged cutting-edge deep learning frameworks like Theano and Caffe to develop deep learning models, including fine-tuning pre-trained models for NLP and image recognition tasks.
- Built robust predictive models and secure APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning). Applied Scikit-learn and Keras for projects focused on pattern discovery and predictive analytics.
- Used AWS infrastructure (EC2, S3, and RDS) to design, build, and maintain scalable systems for hosting machine learning models.
- Integrated databases like PostgreSQL or MongoDB for managing and analyzing data in projects, enabling real-time and context-aware insights.
- Implemented cloud-based architectures with AWS ECS for containerized microservices and REST APIs, ensuring efficient model integration and streamlined data ingestion.
- Developed an intelligent web scraper (Python, requests, BeautifulSoup, pandas) with a simple UI built using Django or Flask to demonstrate robust data acquisition and processing capabilities.
- **Systems Engineering & Automation**
- Utilized GPU-accelerated systems to develop and optimize machine learning models for research projects. Developed and tested algorithmic strategies by applying emerging principles in statistical analysis.
- Designed and developed scalable backend APIs and integrations using Python frameworks (Django, Flask), optimizing data pipelines for efficient machine learning model training and enabling near real-time predictions on datasets.
- Automated the deployment of machine learning applications and services, experimenting with early cloud-native AWS services like EC2, ECS, and Lambda to ensure high availability and streamlined deployment workflows.
- Applied Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (e.g., Jenkins) for secure and continuous application delivery across various environments.
- Automated the creation of secure and scalable infrastructure for machine learning experimentation, enabling robust deployment of Python (Django, Flask) and other server-side applications across cloud platforms like AWS, Azure, and GCP.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to

develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: ML Engineer /Automation Engineer/ Software Test Engineer /DevOps/Data Engineer
Responsibilities:

- **Application Development & Machine Learning**
- Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. Proficient in Python and Java, with capabilities to design, develop, and optimize machine learning applications that deliver actionable insights.
- Developed and deployed predictive models using Python libraries like scikit-learn to accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain.
- Back-End Development: Engineered secure and scalable backend services using Python (e.g., Flask) and Java to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems.
- Front-End Development: Designed and implemented user interfaces with jQuery and JavaScript, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows.
- Extensive expertise in programming languages such as Java, Python, C#, and PowerShell scripting languages.

- **Data Engineering & Architecture**
- Cloud & Data Engineering: Architected and implemented data solutions on AWS, focusing on traditional data warehousing and ETL processes.
- Batch layer: Utilized Amazon EMR, a managed service for running Apache Hadoop clusters, for batch processing of large data volumes.
- Speed layer: Employed AWS Redshift for near-real-time data analysis.
- Designed and developed data solutions using AWS services like EC2, RDS, and S3.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created mapping documents that defined how source data fields corresponded to target data fields.
- Implemented mapping logic based on the mapping document, including writing code to map the source data to the target data according to defined rules.
- **DevOps & Automation**
- Automated deployment of scalable machine learning applications and services, leveraging AWS services like EC2 to ensure high availability.
- Championed continuous integration and delivery (CI/CD) pipelines using tools like Jenkins for secure, continuous application delivery.
- Utilized PowerShell scripting to automate system configurations, testing, and deployment processes.
- Followed software development best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps/Data Engineer /Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Software Development and Deployment: Architected and built scalable web applications using Python and deployed them on traditional servers.
- Infrastructure Management: Managed infrastructure using scripting languages and automation tools.
- Application Development: Developed applications using Python for data processing and business logic.
- Data Management: Utilized relational databases for data storage and retrieval.
- System Integration: Implemented APIs and integrated with diverse systems.
- DevOps and Automation: Deployed and maintained web applications on virtual servers. Managed infrastructure using shell scripts and other automation tools, integrating them into CI/CD pipelines (e.g., Jenkins). Developed REST APIs using Python and Java. Monitored server resources using on-premise tools and implemented logging, monitoring, and security measures.
- Security and Compliance: Implemented security best practices and regularly patched systems and applications.
- Agile & Data Expertise: Experienced working in agile environments with strong data engineering practices.
- Other Proficiencies: Possessed a strong understanding of SQL, Python scripting, and SharePoint.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA

Responsibilities:

- Spearheaded the improvement of continuous integration (CI) and automated deployment processes, leading to a significant reduction in software feature delivery time.
- Utilized scripting languages and configuration management tools to implement Infrastructure as Code (IaC) principles, enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Unix/Linux-based systems, ensuring the smooth operation of critical applications.
- Provided Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, and SharePoint.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running costs.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based system, significantly reducing IT operational costs and integrating with a Salesforce Automation (SFA) platform for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESX & vSphere 4.x, HP/EMC/Dell SAN Storage, and Windows Server 2008, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for desktop productivity suite issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for Business Intelligence (BI) solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized C++, C#, VB.NET, and T-SQL for SSIS and SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure, schema definitions, and Fact and Relational Dimensions tables.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, and SSRS.
- Used SSIS ETL tools to extract and transform complex data, loading it into the data warehouse to produce Analysis Services Cubes for business reporting via SSRS or Data Marts to serve business applications.
- Created and deployed reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting BI solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases, ensuring data accuracy and integrity.
- Defined the target warehouse module, and created target user, module, and schema.
- Built and deployed machine learning models using k-nearest neighbors (k-NN) for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing VBScript and batch files to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos

8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0

- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), LM-Studio, Azure OpenAI ML Services, Copilot, Google GEMINI, MakerSuite
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
  Workflow & Automation Platforms: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Staff Engineer-Python/GenAI | Greenfield AI Build | Enterprise Innovation – EKS-AKS- Paxus – 4/8/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Staff Engineer-Python/GenAI | Greenfield AI Build | Enterprise Innovation /Full Stack Developer/Snr DevOps /Data Engineer / Automation Engineer/Software Engineer/Mulesoft /AI/ML/DL/BI/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Staff Engineer - Python / GenAI | Greenfield AI Build | Enterprise Innovation /Automation Engineer
Responsibilities:

- **AI/ML & Generative AI Solutions**
- Architected and engineered production-grade Generative AI applications by leading the development of Retrieval-Augmented Generation (RAG) pipelines, advanced LLM orchestration, and seamless model integration with enterprise systems, ensuring scalable and reliable AI-driven insights with a single call. response = rag_pipeline.invoke({"query": user_input})
- Developed scalable Python applications, leveraging LangChain (for LLM orchestration) and LlamaIndex (for data indexing and retrieval), Whisper for accurate transcription of high-volume customer calls, Bedrock for accessing foundational models,, and OpenAI frameworks, to integrate GenAI models with real-world data and CRM workflows, and deploying and optimising these systems on AWS while integrating with APIs, microservices, and data platforms.
- Engineered scalable, mission-critical production applications in Python, driving robust system integrations and continuous deployment for AI services with core code such as requests.post(f"{api_url}/api/v1/deployments", json={'app':'genai-service', 'version':'v1.0.0'}).
- Led end-to-end GenAI solution development from secure audio ingestion, automated transcription, LLM orchestration, RAG pipeline integration, to intelligent summarization—ensuring seamless data flow across multi-cloud.
- Developed advanced agentic applications leveraging cutting-edge agentic frameworks including LangGraph, AutoGen, CrewAI, and Semantic Kernel for sophisticated LLM orchestration and autonomous AI agent development.
- Utilized inter-agent communication protocols and seamless integration with external tools (e.g., web search APIs) within these agentic applications, further operationalizing through n8n for workflow automation to drive enhanced operational efficiency, automation, and intelligent decision-making.
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs). retrieved_context = graph_db.query("MATCH (e:Entity)-[r]->(n:Node) WHERE e.name CONTAINS 'query' RETURN e,r,n")
- **AI/ML & Deep Learning Models**
- Leveraged PyTorch, TensorFlow, and Hugging Face Transformers to develop and deploy LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and deep learning models, including fine-tuning pre-trained models for NLP and analytics.
- Built robust predictive models and secure AI APIs using diverse machine learning algorithms (KNN, CNN, RNN, ANN, GAN, regression, decision trees, random forests) and ML paradigms (supervised, unsupervised, reinforcement learning), applying Scikit-learn and Keras to drive Next Best Action (NBA) recommendations and pattern discovery.
- Leveraged a self-recursive Artificial Superintelligence Architecture (ASI-ARCH) on the Harrison Platform (Nvidia DGX-1 systems) to autonomously innovate and optimize AI models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition. emergent_arch = harrison.search(TradingNet, data=multi_modal_feed, objective='SharpeRatio', recursive=True)
- Utilized ASI-ARCH (Artificial Superintelligence Architecture) on Nvidia DGX-1 systems for autonomous innovation in AI models for financial management, enhancing algorithmic trading strategies and risk mitigation.

- Backend, API, and System Integrations
- Designed and developed scalable backend APIs and seamless integrations using Python frameworks (Django, Flask), optimized data pipelines for AI model inference, enabling real-time predictions on large datasets.
- Integrated MongoDB Atlas as a vector database for RAG (Retrieval-Augmented Generation), enabling real-time and context-aware enterprise insights.
- Implemented cloud-native architectures with AWS ECS/EKS for containerized microservices and REST APIs, ensuring efficient model integration with enterprise platforms and streamlined data ingestion within an Amazon Redshift-powered lakehouse.
- Developed an intelligent AI agent using an Amazon Scraper (Python, requests, BeautifulSoup, pandas) with Streamlit UI, demonstrating robust data acquisition and constraints bypassing.
- **Automation, CI/CD, DevOps & Infrastructure-as-Code**
- Automated the deployment of scalable GenAI applications and microservices across hybrid cloud environments, leveraging cloud-native AWS services like ECS/EKS, Lambda, & SageMaker to ensure high availability & streamlined CI/CD pipelines. sagemaker_client.create_endpoint_config(EndpointConfigName='genai-endpoint-config', ProductionVariants=variants)
- Championed Infrastructure-as-Code (Terraform) and automated CI/CD pipelines (GitHub Actions) for secure, continuous GenAI application delivery across complex hybrid cloud environments.
- Automated creation of secure, scalable IaC CI/CD pipelines and isolated GenAI platforms using LLMs (OLLLAMA4.0 on private clouds), MCP+RAG architectures, and agentic n8n workflows, enabling robust deployment of Python (Django, Flask) and Next.js server-side applications across AWS, Azure, and GCP.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Automation Engineer-Python, Ansible /Senior DevOps/Data Engineer /AI/ML/DL/Mulesoft Developer
Responsibilities:

- Spearheaded the design and deployment of scalable microservices architectures using Python & TypeScript on AWS and Azure, integrating advanced ML applications to support trading and compliance platforms for automation and productivity.
- Developed and maintained machine learning pipelines using TensorFlow, Scikit-learn, and Keras, enabling predictive analytics in financial risk forecasting and regulatory data modeling.
- Developed predictive models using machine learning algorithms (e.g., linear regression, logistic regression, decision trees, random forests) with Scikit-learn and Keras, enabling data-driven Next Best Action (NBA) strategies and delivering highly personalized customer experiences (e.g., recommended_action_id = model.predict(customer_profile_features)).
- Applied Natural Language Processing (NLP) with early transformer models such as word2vec and GloVe to automate sentiment analysis of financial news and regulatory document parsing.
- Leveraged a deep reinforcement learning architecture on the NVIDIA DGX-1 platform to autonomously innovate and optimize AI models for financial management, enhancing algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition.
- Proficiently deployed and optimized machine learning models on AWS, leveraging custom-built pipelines on EC2 for comprehensive model lifecycle management, and utilizing serverless Lambda functions for scalable and resilient operations.
- Engineered modular AI components interfacing with SageMaker and early AWS ML services for training, model evaluation, and batch inference.
- Implemented AWS Lambda, S3, and EC2-based batch processing systems to orchestrate real-time ML model execution and data preprocessing tasks at scale.
- **Backend/API Development & Systems Integration**
- Built and exposed robust RESTful APIs for real-time data exchange across core banking systems and regulatory gateways.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.
    1. API Development: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
    2. System Integration: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.
- Engineered scalable, mission-critical backend services and production applications in Python, driving robust system integrations and automated deployments with code such as requests.post(f"{api_url}/v2/deployments", json={'app':'service', 'version':'v1.0'}).
- Designed and implemented cloud-native architectures utilizing containerized microservices on AWS EC2 and REST APIs to facilitate seamless model integration with diverse enterprise data platforms, ensuring efficient data ingestion and processing into a high-performance Amazon Redshift enterprise data warehouse (EDW).
- Optimized financial data workflows by creating robust Python scripts and API integrations, automating interactions between trading infrastructure, databases (e.g., MongoDB), and external financial platforms to enhance operational efficiency and data integrity.
- **Automation, DevOps & Cloud Infrastructure**

- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments.
- E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- Automated resilient infrastructure and real-time application deployments using Ansible and Terraform within Kubernetes/OpenShift clusters, enabling scalable, repeatable workflows for mission-critical financial trading services (e.g., terraform apply -auto-approve).
- Streamlined RBA financial infrastructure through Infrastructure as Code (IaC) practices, utilizing Bicep and Terraform to define and manage consistent infrastructure across diverse ecosystems (e.g., terraform apply -auto-approve).
- Automated environment provisioning and CI/CD for AI pipelines using Jenkins, CloudFormation, and custom Python scripts.
- Accelerated RBA trading operations by designing and implementing automated Jenkins CI/CD pipelines, integrated with GitHub Actions, for rapid, reliable, and secure deployment of trading applications and infrastructure updates across hybrid cloud setups (e.g., kubectl apply -f deployment.yaml).
- Developed and implemented robust CI/CD pipelines using GitHub Actions to streamline Mulesoft application deployment; leveraged YAML scripting and automation to enhance RBA development efficiency and reduce risk.
- Developed robust, data-driven pipelines for automating AWS IaC pipeline creation, prediction, and optimization using SageMaker, CloudFormation, and boto3 (e.g., cfn = boto3.client('cloudformation') …).
- **Cloud/Data Platforms and Portability**
- Designed a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Automation Engineer-Python, Ansible /Senior DevOps/Data Engineer/SharePoint/D365 Developer
Responsibilities:

- Results-oriented Python Application Developer with a strong quantitative background, skilled in creating high-performance, data-driven solutions. Proven expertise in Python and full-stack JavaScript, uniquely positioning me to design, develop, and optimize cutting-edge machine learning applications.
- Developed automated analytical systems using advanced machine learning libraries and data processing pipelines, integrating with core enterprise systems to power intelligent data analysis, information extraction, and automated decision-making in financial services applications.
- Architected cloud-based solutions with event-driven distributed systems on AWS, utilizing EC2, S3, and messaging queues to securely handle high-volume financial transactions while incorporating machine learning-powered insights for fraud detection and risk modeling.
- Proactively researched and evaluated emerging machine learning technologies, leveraging advanced natural language processing (NLP) methods to design and implement intelligent conversational interfaces that enhanced customer experience in banking applications.
- Applied predictive analytics and statistical modeling to forecast international student enrollment rates based on demographic and behavioral data, utilizing custom-built machine learning algorithms (e.g., predicted_students = predict_enrollment(prospective_students.copy())).
- **DevOps / Automation & Cloud Engineering:**
- Engineered and delivered end-to-end automation solutions for student enrolment financial management systems using Python and Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing custom modules and plugins to enable secure, low-latency infrastructure provisioning and configuration across hybrid cloud, on-premises, and RHEL9 environments. (e.g., ansible-playbook deploy_low_latency_trading_engine.yml for seamless deployment workflows)
- Automated robust infrastructure and real-time application deployments through Ansible and Terraform within Kubernetes/OpenShift clusters, guaranteeing scalable, repeatable, and cloud-agnostic workflows for mission-critical financial management services supporting university enrolment. (e.g., terraform apply -auto-approve)
- Enhanced operational efficiency by designing and implementing automated Jenkins CI/CD pipelines, integrated with GitHub Actions, to ensure rapid, dependable, and secure delivery of application and infrastructure updates across hybrid cloud env.
- Implemented AI/ML solutions to optimize software engineering productivity, integrating advanced automation into CI/CD pipelines for faster code validation and testing.
- Managed and optimized cloud-based applications for the student enrollment system, ensuring high availability, scalability, and cost-effectiveness on leading cloud platforms. E.g., aws s3 sync local_data/ s3://uc-student-data-bucket/ --delete
- **Application Security & Optimization**: Secured and optimized critical application functionalities by integrating zero-trust principles and best practices to ensure reliability, scalability, and compliance for sensitive student data. E.g., const helmet = require('helmet'); app.use(helmet());
- Provided technical supporting and troubleshooting cloud resources to ensure availability, resiliency and costing efficiency.
- Software Engineering & Systems Integration

- Built microservices applications using Python and JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.
- Developed intuitive and responsive user interfaces for the UC student application portal using React.js and Redux, enhanced by modern UI libraries for a seamless user experience. E.g., dispatch({ type: 'UPDATE_APPLICANT_STATUS', payload: 'submitted' });
- Cloud Integration Solution Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Enhanced Automation for University IT: Proficient in designing and implementing comprehensive automation solutions with Python and Ansible Tower, including the development of custom modules and plugins, to streamline secure infrastructure provisioning and configuration for the University of Canberra's IT systems, particularly those supporting international student services, across hybrid cloud and RHEL environments. E.g., python migrate_student_data.py
- Resilient System Deployment for Student Services: Orchestrated automated, resilient infrastructure and application deployments leveraging Ansible and Terraform within Kubernetes/OpenShift clusters, guaranteeing scalable and repeatable workflows for vital University services, such as the international student enrolment system, across diverse cloud platforms. E.g., oc apply -f student-api-deployment.yaml

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Python Application Developer /DevOps Engineer/Data Engineer /Software Test Engineer
Responsibilities:

- Data-driven Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. My proficiency in Python and full-stack JavaScript allows me to design, develop, and optimize advanced machine learning applications that deliver actionable insights.
- Applied these capabilities to develop machine learning models that accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain. E.g.: high_risk_claims = claims[model.predict_proba(features)[:, 1] > 0.7]
- Machine Learning Models: Developed and deployed advanced predictive models using Python and Azure Machine Learning, optimizing claims processing efficiency and enhancing fraud detection capabilities. E.g., model_output = trained_model.predict(new_claim_features)
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages.
- **Front-End & Back-End Development:**
- Front-End Development: Designed and implemented intuitive web interfaces with Next.js, streamlining the user experience for complex forms and dashboards crucial to efficient injury compensation claim management.
- E.g., const { data: claimData } = await useSWR('/api/claim-details');
- Front-End Development (React.js & Redux): Developed intuitive and responsive user interfaces using React.js and Redux, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows. E.g., dispatch(setClaimStatus('Submitted'));
- Back-End Development (Node.js): Engineered secure and scalable Node.js backend services to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems. E.g., app.post('/api/claims', async (req, res) => { /* secure claim intake */ });
- **Cloud Architecture & Data Engineering:** Architected and implemented a modern 3-layer lambda architecture on AWS:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed AWS Redshift with its built-in data processing capabilities for near-real-time processing & analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- **DevOps, MLOps & Team Enablement**
- Set up Kubernetes Cluster by creating a Kubernetes cluster using GKE, EKS, AKS, or local deployment tools like Minikube. Ensured the cluster had sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Selected suitable MLOps frameworks/platforms that integrate with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core, or Argo Workflows, and followed MLOps best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**

Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilized AWS Cloud Development Kit (CDK) for infrastructure provisioning and management.
- Developed serverless applications using AWS Lambda; worked with Amazon DynamoDB for data storage and retrieval.
- Implemented APIs and integrated with diverse systems using AWS API Gateway.
- Deployed and maintained web applications on Amazon Web Services (AWS) cloud platform.
- Managed infrastructure as code (IaC) with Terraform, integrating tools into CI/CD pipelines (Jenkins, GitLab CI/CD), developed REST APIs (Python, Java), monitored AWS resources (EC2, S3) using CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Security and Compliance: Implemented security best practices; regularly patched systems and applications.
- Agile & Data Expertise: Experienced working in agile environments with strong data engineering practices.
- GCP Stack Proficiency: Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages.

Aug.2009 - Oct. 2011 Manulife Financial Corporation (MLJ), Tokyo
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IaC), enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, Office 365, SharePoint, and more.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 ING Life Insurance Company, LTD, Tokyo
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based ING Link project, significantly reducing IT operational costs and integrating with Salesforce Automation (SFA) cloud for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for Office Online issues.

Jul. 2004 to Apr. 2008 Microsoft Japan, Tokyo
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure landscape, schema definitions, Fact and Relational Dimensions tables, Extended Star Schema (hierarchies, attributes, and text), Wormholes, Constellation, and Galaxy schemas with Multiple Fact Tables Sharing Dimensions using MDX and BIML script.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, & SSRS.
- Using SSIS ETL Tools to extract complex data into the DW staging area, transformed and loaded cleansed data as Master Data for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Marts to serve business apps.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting MS BI Solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Defined Target Warehouse Module, created Target User, Target Module, and Target schema.
- Built and deployed ML models using k-nearest neighbors (k-NN), non-parametric supervised learning methods, and Python for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing PowerShell scripts to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), LM-Studio, Azure OpenAI ML Services, Copilot, Google GEMINI, MakerSuite
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
  Workflow & Automation Platforms: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve

Staff Engineer-Python/GenAI | Greenfield AI Build | Enterprise Innovation –EKS –AKS – Bluefin- 3/8/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Staff Engineer-Python/GenAI | Greenfield AI Build | Enterprise Innovation /Full Stack Developer/Snr DevOps /Data Engineer / Automation Engineer/Software Engineer/Mulesoft /AI/ML/DL/BI/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Staff Engineer - Python / GenAI | Greenfield AI Build | Enterprise Innovation /Automation Engineer
Responsibilities:

- **Generative AI & Advanced AI/ML Solutions**
- Significantly enhanced software engineering productivity by architecting and implementing advanced Generative AI (GenAI) and AI/ML applications via robust Python-based microservices architectures across AWS and Azure cloud platforms. A key component involved integrating GenAI models seamlessly with real-world data and CRM workflows using Bedrock (including OpenAI frameworks), alongside APIs and data platforms, to streamline deployment and optimization.

- Developed scalable Python applications, orchestrating LangChain and LlamaIndex for LLM orchestration and data indexing. Integrated Amazon Transcribe Call Analytics for automatic high-volume call transcription and sentiment analysis, directly logging customer sentiment into CRM workflows.
- Developed advanced agentic applications leveraging cutting-edge agentic frameworks including LangGraph, AutoGen, CrewAI, and Semantic Kernel for sophisticated LLM orchestration and autonomous AI agent development. Utilized inter-agent communication protocols and seamless integration with external tools (e.g., web search APIs) within these agentic applications, further operationalizing through n8n for workflow automation to drive enhanced operational efficiency, automation, and intelligent decision-making.
- Leveraged AWS Kiro's spec-driven development & agentic capabilities to design, build, and maintain scalable, secure AWS infrastructure (EC2, EKS, S3, RDS, Lambda, SageMaker, Amazon Neptune) for hosting advanced Graph RAG pipelines, vector databases, & LLMs, Using AWS Bedrock for diverse foundational models (e.g., Amazon, Anthropic, AI21 Labs). retrieved_context = graph_db.query("MATCH (e:Entity)-[r]->(n:Node) WHERE e.name CONTAINS 'query' RETURN e,r,n")
- **AI/ML & Deep Learning Models**
- Leveraged models like PyTorch, TensorFlow, and Hugging Face Transformers to develop & deploy varied LLMs (BERT, GPT-4o, Jurassic-1, Bloom) & deep learning models. Applied fine-tuning of pre-trained models (BERT, GPT) & integrated various ML/DL algorithms for NLP projects and advanced analytical tasks, delivering accurate, personalized insights.
- Built robust predictive models and secure AI APIs utilizing a range of machine learning algorithms (including KNN, CNN, RNN, ANN, GAN, linear regression, logistic regression, decision trees, random forests) and ML paradigms (unsupervised, supervised, reinforcement learning). Applied Scikit-learn and Keras to drive Next Best Action (NBA) recommendations, enhance customer personalization, and facilitate pattern discovery.
- Leveraged a self-recursive Artificial Superintelligence Architecture (ASI-ARCH) on the Harrison Platform (Nvidia DGX-1 systems) to autonomously innovate and optimize AI models for financial management. Enhanced algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition. emergent_arch = harrison.search(TradingNet, data=multi_modal_feed, objective='SharpeRatio', recursive=True)
- **Backend, API, and System Integrations**
- Designed and developed scalable and performant applications with robust backend APIs and seamless system integrations using Python frameworks (e.g., Django, Flask), including optimized data pipelines for AI model inference that enabled real-time predictions from large datasets.
- **Cloud Infrastructure & Specialized Tools**: Integrated MongoDB Atlas as a vector database for Retrieval-Augmented Generation (RAG) solutions, enabling real-time, context-aware insights from enterprise search services.
- Designed and implemented cloud-native architectures employing AWS ECS/EKS for containerized microservices and REST APIs to facilitate seamless model integration with diverse enterprise data platforms, ensuring efficient data ingestion and processing within an Amazon Redshift-powered data lakehouse.
- Developed an intelligent AI agent utilizing an Amazon Scraper built with Python (leveraging requests and BeautifulSoup, pandas for data handling) and Streamlit for the user interface, demonstrating capabilities to bypass restrictions.
- **Automation, CI/CD, DevOps & Infrastructure**
- Automation, CI/CD, & DevOps: Automated the creation of secure, scalable Infrastructure-as-Code (IaC) CI/CD pipelines, built on secure, isolated Generative AI platforms utilizing LLMs (self-hosted models deployed on OLLAMA4.0 within secure private virtual networks), MCP+RAG architectures, and n8n agentic workflows—to enable robust deployment of Python web applications (Django, Flask) and high-performance Next.js solutions with server-side rendering across hybrid cloud environments (Azure, AWS, GCP). E.g., curl -X POST https://n8n.yourdomain.com/webhook/generate-cicd-pipeline
- Designed and implemented end-to-end automation solutions using Python and Ansible (Tower, AWX, Red Hat AAP 2.5), developing custom modules and plugins to streamline infrastructure provisioning and configuration across hybrid cloud, on-prem, and Red Hat Enterprise Linux (RHEL9/10) environments. E.g., ansible-playbook provision_rhel_app_servers.yml
- Automated infrastructure and application deployments using Ansible and Terraform in Kubernetes/OpenShift clusters, ensuring scalable and repeatable workflows across cloud-agnostic platforms.
  E.g., terraform apply -auto-approve && ansible-playbook deploy_app_on_openshift.yml
- Built automated Jenkins pipelines leveraging Jenkinsfile Runner via GitHub Actions to support comprehensive automated testing, robust deployment, and infrastructure updates in multi-environment hybrid cloud setups.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Automation Engineer-Python, Ansible, CICD, APIs /Snr DevOps/Data Engineer/Mulesoft Developer
Responsibilities:

- Spearheaded the design and deployment of scalable microservices architectures using Python & TypeScript on AWS and Azure, integrating advanced ML applications to support trading and compliance platforms for automation and productivity.
- Developed and maintained machine learning pipelines using TensorFlow, Scikit-learn, and Keras, enabling predictive analytics in financial risk forecasting and regulatory data modeling.

- Developed predictive models using machine learning algorithms (e.g., linear regression, logistic regression, decision trees, random forests) with Scikit-learn and Keras, enabling data-driven Next Best Action (NBA) strategies and delivering highly personalized customer experiences (e.g., recommended_action_id = model.predict(customer_profile_features)).
- Applied Natural Language Processing (NLP) with early transformer models such as word2vec and GloVe to automate sentiment analysis of financial news and regulatory document parsing.
- Leveraged a deep reinforcement learning architecture on the NVIDIA DGX-1 platform to autonomously innovate and optimize AI models for financial management, enhancing algorithmic trading strategies and risk mitigation by discovering emergent market principles beyond human intuition.
- Proficiently deployed and optimized machine learning models on AWS, leveraging custom-built pipelines on EC2 for comprehensive model lifecycle management, and utilizing serverless Lambda functions for scalable and resilient operations.
- Engineered modular AI components interfacing with SageMaker and early AWS ML services for training, model evaluation, and batch inference.
- Implemented AWS Lambda, S3, and EC2-based batch processing systems to orchestrate real-time ML model execution and data preprocessing tasks at scale.
- **Backend/API Development & Systems Integration**
- Built and exposed robust RESTful APIs for real-time data exchange across core banking systems and regulatory gateways.
- Developed and deployed scalable applications using MuleSoft Anypoint Platform, including Mule 4 & Anypoint Studio 7.x, with a strong focus on robust backend API development and seamless system integrations.
    1. API Development: POST /api/v1/orders HTTP/1.1 { "item": "widget", "qty": 2 }
    2. System Integration: Flow orchestrates Salesforce, SAP, and custom REST services for order processing.
- Engineered scalable, high-performance Python applications and robust backend APIs, facilitating seamless cloud-native integrations across Azure/AWS services for real-time data processing and AI solution enablement.
- Designed and implemented cloud-native architectures utilizing containerized microservices on AWS EC2 and REST APIs to facilitate seamless model integration with diverse enterprise data platforms, ensuring efficient data ingestion and processing into a high-performance Amazon Redshift enterprise data warehouse (EDW).
- Optimized financial data workflows by creating robust Python scripts and API integrations, automating interactions between trading infrastructure, databases (e.g., MongoDB), and external financial platforms to enhance operational efficiency and data integrity.
- **Automation, DevOps & Cloud Infrastructure**
- Specialized in designing and implementing end-to-end automation solutions using Python & Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing modules and plugins to streamline secure, low-latency infrastructure provisioning for RBA trading platforms across hybrid cloud, on-prem, and RHEL9 environments.
- E.g., ansible-playbook deploy_low_latency_trading_engine.yml to automate trading engine deployments.
- Automated resilient infrastructure and real-time application deployments using Ansible and Terraform within Kubernetes/OpenShift clusters, enabling scalable, repeatable workflows for mission-critical financial trading services (e.g., terraform apply -auto-approve).
- Streamlined RBA financial infrastructure through Infrastructure as Code (IaC) practices, utilizing Bicep and Terraform to define and manage consistent infrastructure across diverse ecosystems (e.g., terraform apply -auto-approve).
- Automated environment provisioning and CI/CD for AI pipelines using Jenkins, CloudFormation, and custom Python scripts.
- Accelerated RBA trading operations by designing and implementing automated Jenkins CI/CD pipelines, integrated with GitHub Actions, for rapid, reliable, and secure deployment of trading applications and infrastructure updates across hybrid cloud setups (e.g., kubectl apply -f deployment.yaml).
- Developed and implemented robust CI/CD pipelines using GitHub Actions to streamline Mulesoft application deployment; leveraged YAML scripting and automation to enhance RBA development efficiency and reduce risk.
- Developed robust, data-driven pipelines for automating AWS IaC pipeline creation, prediction, and optimization using SageMaker, CloudFormation, and boto3 (e.g., cfn = boto3.client('cloudformation') …).
- **Cloud/Data Platforms and Portability**
- Designed a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Automation Engineer-Python, Ansible, CICD, APIs /Snr DevOps/Data Engineer/SharePoint/D365 Developer
Responsibilities:

- Results-oriented Python Application Developer with a strong quantitative background, skilled in creating high-performance, data-driven solutions. Proven expertise in Python and full-stack JavaScript, uniquely positioning me to design, develop, and optimize cutting-edge machine learning applications.

- Developed automated analytical systems using advanced machine learning libraries and data processing pipelines, integrating with core enterprise systems to power intelligent data analysis, information extraction, and automated decision-making in financial services applications.
- Architected cloud-based solutions with event-driven distributed systems on AWS, utilizing EC2, S3, and messaging queues to securely handle high-volume financial transactions while incorporating machine learning-powered insights for fraud detection and risk modeling.
- Proactively researched and evaluated emerging machine learning technologies, leveraging advanced natural language processing (NLP) methods to design and implement intelligent conversational interfaces that enhanced customer experience in banking applications.
- Applied predictive analytics and statistical modeling to forecast international student enrollment rates based on demographic and behavioral data, utilizing custom-built machine learning algorithms (e.g., predicted_students = predict_enrollment(prospective_students.copy())).
- **DevOps / Automation & Cloud Engineering:**
- Engineered and delivered end-to-end automation solutions for student enrolment financial management systems using Python and Red Hat Ansible Automation Platform (including Ansible Tower and AWX), developing custom modules and plugins to enable secure, low-latency infrastructure provisioning and configuration across hybrid cloud, on-premises, and RHEL9 environments. (e.g., ansible-playbook deploy_low_latency_trading_engine.yml for seamless deployment workflows)
- Automated robust infrastructure and real-time application deployments through Ansible and Terraform within Kubernetes/OpenShift clusters, guaranteeing scalable, repeatable, and cloud-agnostic workflows for mission-critical financial management services supporting university enrolment. (e.g., terraform apply -auto-approve)
- Enhanced operational efficiency by designing and implementing automated Jenkins CI/CD pipelines, integrated with GitHub Actions, to ensure rapid, dependable, and secure delivery of application and infrastructure updates across hybrid cloud env.
- Implemented AI/ML solutions to optimize software engineering productivity, integrating advanced automation into CI/CD pipelines for faster code validation and testing.
- Managed and optimized cloud-based applications for the student enrollment system, ensuring high availability, scalability, and cost-effectiveness on leading cloud platforms. E.g., aws s3 sync local_data/ s3://uc-student-data-bucket/ --delete
- **Application Security & Optimization**: Secured and optimized critical application functionalities by integrating zero-trust principles and best practices to ensure reliability, scalability, and compliance for sensitive student data. E.g., const helmet = require('helmet'); app.use(helmet());
- Provided technical supporting and troubleshooting cloud resources to ensure availability, resiliency and costing efficiency.
- Software Engineering & Systems Integration
- Built microservices applications using Python and JavaScript with frameworks like Flask & Node.js. Expertise in consuming RESTful web services to integrate with UC's enterprise services layer, databases, and host systems.
- Developed intuitive and responsive user interfaces for the UC student application portal using React.js and Redux, enhanced by modern UI libraries for a seamless user experience. E.g., dispatch({ type: 'UPDATE_APPLICANT_STATUS', payload: 'submitted' });
- Cloud Integration Solution Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Enhanced Automation for University IT: Proficient in designing and implementing comprehensive automation solutions with Python and Ansible Tower, including the development of custom modules and plugins, to streamline secure infrastructure provisioning and configuration for the University of Canberra's IT systems, particularly those supporting international student services, across hybrid cloud and RHEL environments. E.g., python migrate_student_data.py
- Resilient System Deployment for Student Services: Orchestrated automated, resilient infrastructure and application deployments leveraging Ansible and Terraform within Kubernetes/OpenShift clusters, guaranteeing scalable and repeatable workflows for vital University services, such as the international student enrolment system, across diverse cloud platforms. E.g., oc apply -f student-api-deployment.yaml

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Python Application Developer /DevOps Engineer/Data Engineer /Software Test Engineer
Responsibilities:

- Data-driven Python Application Developer with a robust quantitative foundation, adept at building high-performance solutions. My proficiency in Python and full-stack JavaScript allows me to design, develop, and optimize advanced machine learning applications that deliver actionable insights.
- Applied these capabilities to develop machine learning models that accurately assess the likelihood of specific claim outcomes or potential fraud indicators, enabling proactive risk management and optimized resource allocation in the injury compensation domain. E.g.: high_risk_claims = claims[model.predict_proba(features)[:, 1] > 0.7]
- Machine Learning Models: Developed and deployed advanced predictive models using Python and Azure Machine Learning, optimizing claims processing efficiency and enhancing fraud detection capabilities. E.g., model_output = trained_model.predict(new_claim_features)
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages.

- **Front-End & Back-End Development:**
- Front-End Development: Designed and implemented intuitive web interfaces with Next.js, streamlining the user experience for complex forms and dashboards crucial to efficient injury compensation claim management.
- E.g., const { data: claimData } = await useSWR('/api/claim-details');
- Front-End Development (React.js & Redux): Developed intuitive and responsive user interfaces using React.js and Redux, optimizing the claimant experience for EML's injury compensation portal and streamlining data presentation for complex claim workflows. E.g., dispatch(setClaimStatus('Submitted'));
- Back-End Development (Node.js): Engineered secure and scalable Node.js backend services to efficiently manage high volumes of injury compensation claim data, ensuring data integrity and seamless integration with critical EML systems. E.g., app.post('/api/claims', async (req, res) => { /* secure claim intake */ });
- **Cloud Architecture & Data Engineering:** Architected and implemented a modern 3-layer lambda architecture on AWS:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed AWS Redshift with its built-in data processing capabilities for near-real-time processing & analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.
- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- **DevOps, MLOps & Team Enablement**
- Set up Kubernetes Cluster by creating a Kubernetes cluster using GKE, EKS, AKS, or local deployment tools like Minikube. Ensured the cluster had sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Selected suitable MLOps frameworks/platforms that integrate with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core, or Argo Workflows, and followed MLOps best practices for versioning, deployment, testing, and automation.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilized AWS Cloud Development Kit (CDK) for infrastructure provisioning and management.
- Developed serverless applications using AWS Lambda; worked with Amazon DynamoDB for data storage and retrieval.
- Implemented APIs and integrated with diverse systems using AWS API Gateway.
- Deployed and maintained web applications on Amazon Web Services (AWS) cloud platform.
- Managed infrastructure as code (IaC) with Terraform, integrating tools into CI/CD pipelines (Jenkins, GitLab CI/CD), developed REST APIs (Python, Java), monitored AWS resources (EC2, S3) using CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Security and Compliance: Implemented security best practices; regularly patched systems and applications.
- Agile & Data Expertise: Experienced working in agile environments with strong data engineering practices.
- GCP Stack Proficiency: Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IaC), enhancing deployment processes and scalability.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures, including Active Directory (AD), DNS, DHCP, Exchange 2010, Office 365, SharePoint, and more.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Served as Project Manager to replace the existing insurance web application (from IBM AS/400 and LAMP) with a WISA-based ING Link project, significantly reducing IT operational costs and integrating with Salesforce Automation (SFA) cloud for improved marketing results.
- Planned, implemented, and supported all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, and applied security principles and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Designed data warehouse architecture, including hardware infrastructure landscape, schema definitions, Fact and Relational Dimensions tables, Extended Star Schema (hierarchies, attributes, and text), Wormholes, Constellation, and Galaxy schemas with Multiple Fact Tables Sharing Dimensions using MDX and BIML script.
- Developed data warehouse and business reporting solutions to meet stakeholder requirements using SSIS, SSAS, & SSRS.
- Using SSIS ETL Tools to extract complex data into the DW staging area, transformed and loaded cleansed data as Master Data for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Marts to serve business apps.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio, including debugging, monitoring, and troubleshooting MS BI Solutions.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Defined Target Warehouse Module, created Target User, Target Module, and Target schema.
- Built and deployed ML models using k-nearest neighbors (k-NN), non-parametric supervised learning methods, and Python for customer churn prediction and fraud detection, achieving a 15% improvement in accuracy.
- Involved in UAT and TEST in a QA development environment by writing PowerShell scripts to automate processes.

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Languages & Frameworks:** Java, JavaScript (including JQuery, jqGrid), C#, VB/VBA, AJAX, ASP.NET (Web Forms, MVC 2~6), ActionScript, XML, T-SQL, PL/SQL, SQL*Plus, ADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, .NET Framework/Core (2.0~6.0), AvePoint, K2 Black Pearl
- **OS**: Windows, Unix/Linux/BSD (Ubuntu 16~21, RHEL 3~10, SuSE 7~10, CentOS 5.0), Solaris 8~10, HP-UX 8~11, AIX 5.3/6.1
- **Databases & Data Platforms**: Oracle 9i/10gR2/11gR2/12c, Teradata, Exadata, Microsoft SQL Server 2005~2022, Azure SQL, Sybase, MySQL, Power BI
- **Cloud Platforms & Big Data**: Microsoft Azure, AWS Cloud (Amazon Athena, Redshift/Redshift Spectrum), Google Cloud Platform (GCP), Apache Spark, PySpark
- **Development & Data Tools / IDEs**: Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer, SQL Server Management Studio (SSMS) / BIDS, OBIEE 11g~12c, OWB 10gR2~11R2, Business Objects, Cognos 8~11, TM1 10.2, RStudio, R (3.5~4.3), Python (3.7~3.11), PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, Salesforce, MuleSoft Anypoint 7 and Mule 4.0
- **AI, Machine Learning, Deep Learning & BI Platforms**: Harrison Platform on Nvidia DGX-1 to DGX A100 systems, OpenAI (ChatGPT-4o, Multi-GPTs, Auto-Agents), LM-Studio, Azure OpenAI ML Services, Copilot, Google GEMINI, MakerSuite
- **Content Management Systems (CMS) & CRM**: SharePoint 2007~2022, CRM 3.0~D365, AX 2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco 4.x/6.x
- **Servers & Web/Application Servers**: Wintel Servers, Solaris Servers, Linux Servers, GlassFish, Apache Tomcat 6~10.1, WebSphere Application Server (WAS) / MQ, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization & Containerization**: Citrix Xen 5.0/5.5, VMware ESX 3.5~5/ESXi 5.5, IBM PowerVM, TM1,Hyper-V, Sun,Dell VIS
- **Security Technologies**: Microsoft Defender Suite, FortiGate Firewalls, FortiClient, Multi-Factor Authentication (MFA), SSL-VPN, IPSec VPN, Public Key Infrastructure (PKI), Digital Signing (RSA Data Security Inc.), Public Key Cryptographic Standards (PKCS#1), Hash functions (SHA-1, MD5)
  Workflow & Automation Platforms: K2 Workflow, Nintex 2010 Workflow, SharePoint 2007~2022 Workflows, AvePoint DocAve